

Course Code	@21IPC501 J	Course Name	Research Methodology and Publication Ethics	Course Category	C		L	T	P	C
							2	1	2	4

Pre-requisite Courses	Nil	Co-requisite Courses	Nil	Progressive Courses	Nil
Course Offering Department	College of Engineering and Technology	Data Book / Codes/Standards	Statistical data book and tables		

Course Learning Rationale (CLR):	The purpose of learning this course is to:
CLR-1:	Understand the different methodologies involved in research studies
CLR-2:	Familiarize with various types of academic writing and presentations
CLR-3:	Understand the significance of ethics in academic publications
CLR-4:	Familiarize with different design of experiments and optimization techniques
CLR-5:	Familiarize with approaches of analysing data

Course Outcomes (CO):	At the end of this course, learners will be able to:	Programme Outcomes (PO)		
		1	2	3
CO-1:	Acquire knowledge on basic research methodology	3	3	-
CO-2:	Write and present research contents	3	3	-
CO-3:	Recognise the values and ethic involved in scientific research	3	3	-
CO-4:	Apply different approaches for solving design of experiments and optimization problems	3	2	-
CO-5:	Analyse the applications of statistical and machine learning tools for data analysis	3	2	-

Module-1 – RESEARCH PREPARATION, PLANNING AND RESOURCES	16 Hours
Research – objectives, definitions – knowledge economy; Perspectives of stakeholders; Types of research; Critical & creative thinking; Generating research topic; Identifying broad research question and objectives – design of questionnaire; Characteristics of a good research – research design; Methods of scientific enquiry – theoretical, experimental and empirical; Hypothesis – formulation & testing. Sources of information – ICT enabled tools; Databases, repositories, public and private sources, indexes; Literature search – keyword; Backward and forward search; Quality measurement tools: journal metrics, author-level metrics; Meta-analysis of research findings; Literature review: grouping, analysing & comparison: Identification of research Methods; Intellectual Property Rights; Copyright – patent –geographical indication – industrial designs – technology transfer. PRACTICE: Identifying research topic and formulation of Research Statement; Conduction of literature review.	
Module-2 – ACADEMIC WRITING AND PRESENTATIONS	14 Hours
Preparation of project proposal for funding – Identification of funding agencies; Project proposals structure; Research report writing; Data visualization; Communication model – audience analysis; Identification of suitable journal; Conference presentation types: Oral, poster – difference in audience interaction – guideline for effective presentation; Synopsis – thesis –	

extended abstract – graphical and video abstract – short communications; Referencing – style – tools for referencing; Thesis writing: structure – preliminary pages, main body, references; Evaluation of Thesis – examiner reports – Oral defence. PRACTICE: Writing abstract and conclusion; Referencing in different styles.	
Module-3 – PUBLICATION ETHICS AND MISCONDUCTS	12 Hours
Ethics: definition; Ethics with respect to science and research - intellectual honesty and research integrity; Scientific misconducts: Falsification, Fabrication and Plagiarism (FFP); Redundant publications: duplicate and overlapping publications, salami slicing; Selective reporting and misrepresentation of data; Publication ethics: introduction and importance; Standards setting initiatives and guidelines: COPE, WAME; Conflicts of interest. Publication misconduct: definition, concept, unethical behaviour; Violation of publication ethics, authorship and contributor ship; Identification, complaints and appeals; Predatory publisher and journals. PRACTICE: Plagiarism checking and rectification.	
Module-4 – DESIGN OF EXPERIMENTS AND OPTIMIZATION METHODS	16 Hours
Design of experiments; Basic experimental designs; Completely randomized design; Randomized block design; Latin square design; Full factorial design - 2^2 , 2^3 and 2^k ; Fractional factorial designs; Taguchi methodology – Orthogonal array; S/N ratio – Taguchi method for optimization; Response surface method – multi response optimization. Global optimization techniques – evolutionary and swarm algorithms – examples and case studies. PRACTICE: Design of experiment – Taguchi method – Response surface method.	
Module-5 – DATA ANALYSIS	17 Hours
Basic probability distributions – Normal distribution – Binominal and Poisson distributions – Weibull and exponential distributions; Sampling: types - size, Sampling tests: T-test – F-test – χ^2 test; Modelling – validation and verification; Simulation – techniques; ANOVA; Correlation analyses: Pearson and Spearman Correlations; Regression, classification and cluster analyses; Factor analysis; Discriminant analysis; Uncertainty and error analyses; Machine learning: supervised and unsupervised – examples and case studies. PRACTICE: Distribution; Sampling; Machine learning tools.	

Learning Resources	1. Ganesan, R. (2020), Research Methodology for Engineers, MJP Publishers, Chennai. 2. Kothari, C.R. & Garg, G. (2022), Research Methodology, New Age International Publishers, New Delhi. 3. Singh, U.P., Ahlawat, S. & Sharma, S. (2023), Research and Publication Ethics, S.Chand & Co. Ltd., New Delhi. 4. Mukerjee, S.P. (2020), A Guide to Research Methodology: An Overview of Research Problems, Tasks and Methods, CRC Press, New York. 5. Bairagi, V. & Munot, M.V. (2019), Research Methodology: A practical and scientific approach, CRC Press, New York. 6. Locharoenrat, K. (2017), Research Methodology for beginners, CRC Press, New York. 7. Panneerselvam, R. (2014) Research Methodology, R., PHI Learning Pvt Ltd, New Delhi. 8. Upagade, V. & Shende, A. (2010), A. Research Methodology, Vijay, S.Chand & Co. Ltd., New Delhi. 9. Alvi, M.H. (2016), A Manual for Referencing styles in Research, University of Karachi, Pakistan.	10. Steneck, N.H. (2007). Introduction to the Responsible Conduct of Research. Office of Research Integrity. Available at: https://ori.hhs.gov/sites/default/files/rcrintro.pdf 11. Adams, K.A., and Lawrence, E.K. (2019), Research Methods, Statistics and applications, Sage Publication, Thousand ORKS, California, USA 12. Gastel, B., and Day, R.A. (2016), How to Write and Publish a Scientific paper, 8th Edition, Greenwood, California 13. Kornuta, H.M., and Germaine, R.W., (2006), A concise guide to writing a Thesis or Dissertation, 2nd Edition, Taylor & Francis Group, New York 14. Montgomery, D.C., (2013), Design and Analysis of Experiments, 8 th edition, Wiley, New Delhi. 15.. Box, G.E.P. and Draper N.R, (2007) Empirical Model-Building and Response Surfaces, John Wiley & Sons. 16.. Antony, J., (2014), Design of Experiments for Engineers and Scientists, Second Edition, Elsevier, Amsterdam. 17. Das, M.N. and Giri, N.C., (2003) Design and Analysis of Experiments, New Age International (P) Limited, New Delhi. 18. Chopra, R. (2018), Machine Learning, Khanna Book Publishing, New Delhi.
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Learning Assessment							
	Bloom's Level of Thinking	Continuous Learning Assessment (CLA)				Summative Final Examination (40% weightage)	
		Formative CLA-1 Average of unit test (45%)		Life-Long Learning CLA-2 (15%)			
		Theory	Practice	Theory	Practice	Theory	Practice
Level 1	Remember	20%	-	-	10%-	20%	-

Level 2	<i>Understand</i>	20%	-	-	10%-	20%	-
Level 3	<i>Apply</i>	30%	-	-	40%-	30%	-
Level 4	<i>Analyze</i>	30%	-	-	40%-	30%	-
Level 5	<i>Evaluate</i>	-	-	-	-	-	-
Level 6	<i>Create</i>	-	-	-	-	-	-
	<i>Total</i>	100 %		100 %		100 %	

Course Designers		
Experts from Industry	Experts from Higher Technical Institutions	Internal Experts
1.	1.	1.
2.	2.	2.